

Mohamed Rissal Hedna

Applied Scientist / Research Engineer · LLMs, Calibration, Safety & Evaluation
Hamburg, Germany · +49 176 43402970 · rissalhedna@gmail.com

[GitHub](#) · [Google Scholar](#) · [Portfolio](#)

SUMMARY

Applied Scientist / Research Engineer focused on LLM calibration, safety, and evaluation. First author of SECL (arXiv 2026, under review at EMNLP 2026), with production experience building, leading, and shipping large-scale generative and multi-agent systems.

CORE SKILLS

Research: calibration, uncertainty quantification, hallucination detection, evaluation & benchmarking, SFT, reward modeling, test-time training

ML & Retrieval: PyTorch, Hugging Face, RAG, dense retrieval (Qdrant, LlamaIndex), multi-agent pipelines, synthetic data

Languages & Systems: Python, Java, C, JavaScript/TypeScript, SQL; Docker, CI/CD, Azure, GCP

PUBLICATIONS

- M. R. Hedna, J. Strich, M. Semmann, C. Biemann. “SECL: Self-Calibrating Language Models via Test-Time Discriminative Distillation.” [arXiv:2604.09624](#) [cs.CL], 2026. Under review, EMNLP 2026. [\[code\]](#)
- M. R. Hedna, Y. Djemmal, K. Mershad. “A Model for the Automatic Mixing of Multiple Video and Audio Clips.” Intl. Conference on Cyberworlds (CW), 2023. [\[IEEE Xplore\]](#)
- S. Nder, M. R. Hedna. “Simulated Adversarial Patch Attacks on Vision-Based Logistics Systems.” [arXiv:2511.19254](#), 2024.

RESEARCH EXPERIENCE

Graduate Researcher | University of Hamburg

2025–Present

- Built SECL, a test-time training pipeline that calibrates LLM confidence from the model's own P(“True”) signal on synthetic data, with no labels or human supervision.
- Improved calibration over verbalized-confidence baselines without their inference cost or distribution-shift fragility.

Visiting Research Assistant | Advanced Telerobotics Lab, USA

2022

- Built a custom C/Python CNN reaching sub-50ms inference on embedded NVIDIA Jetson for obstacle detection in GPS-denied environments.
- Implemented LiDAR-vision sensor fusion under tight memory and power constraints on edge devices.

PROFESSIONAL EXPERIENCE

PHAROS Labs | AI Engineer

Hamburg · 2024–Present

- Designed and shipped lightweight hallucination-detection layers for customer-facing LLM features, cutting unsupported answers 78%.
- Ran weekly customer sessions and shipped the resulting fixes end to end across frontend, backend, and MLOps.
- Built automated evaluation and monitoring for reasoning- and retrieval-heavy systems, cutting end-to-end latency 55%.

Fiindo | AI Engineer (Contract)

Hamburg · 2025

- Built the full agentic pipeline solo, converting raw financial data into finished short-form videos across ingestion, narration, MANIM chart animation, and render.
- Led the team building the visual and chart-generation layers on top of the architecture I designed.
- Built a caching and reuse layer that skips unchanged segments, making repeat and near-duplicate renders ~3x faster.

AdaLab UG | Machine Learning Developer

Hamburg · 2023–2024

- Raised peak-load throughput 60% on a multimodal (text, image, audio, video) generation platform via parallel execution and memory profiling.
- Shipped and stabilized production video generation for 500+ active users.

SELECTED PROJECTS

T5-Efficient

PyTorch, Transformers, BF16

- Trained T5 from scratch for git commit message generation.
- Cut parameters 81% (17.7M to 3.3M) via pruning and architecture optimization, converging in under 12h on one GPU.

UniLLM

Qdrant, LlamaIndex, Python

- Built a dense-retrieval RAG assistant with async streaming and an automated crawler for German study and living data.
- Cut query resolution time 13% via optimized chunking and indexing.

EDUCATION

M.Sc. Intelligent Adaptive Systems | University of Hamburg

CGPA 1.69 (German scale, 1.0 = best). Thesis: SECL: Self-Calibrating Language Models. Focus: neural networks, knowledge processing, statistical methods in language technology.

B.Sc. Computer Science | Lebanese American University

CGPA 3.91/4.0, High Distinction. MEPI TL Full Scholarship (competitive merit award).

LANGUAGES

English (C2) · French (C1) · German (B2) · Arabic (Native)